

STAT 516

Covariance and Correlation. Multivariate Case. Multinomial distribution

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April 22, 2021

Motivation

- ▶ For non-independent variables X and Y
 $Var(X + Y) \neq Var(X) + Var(Y)$
- ▶ Instead, we have

$$\begin{aligned} Var(X + Y) &= E(X + Y)^2 - [E(X + Y)]^2 = \dots \\ &= Var(X) + Var(Y) + 2[E(XY) - E(X)E(Y)] \end{aligned}$$

- ▶ Let X and Y be two random variables defined on the common sample space Ω
- ▶ It is assumed that $E(X)$, $E(Y)$ and $E(XY)$ all exist
- ▶ The covariance of X and Y is defined as

$$Cov(X, Y) = E(XY) - E(X)E(Y) = E[(X - E(X))(Y - E(Y))]$$

Interpretation

- ▶ Covariance is a measure of whether two random variables X and Y tend to increase or decrease together
- ▶ For example, taller people tend to weigh more than shorter people; thus, height and weight usually have a positive covariance
- ▶ Covariance is scale dependent and can take arbitrary positive and negative values
- ▶ Renormalization is necessary to make it easier to interpret

Correlation

- ▶ Let X and Y be two random variables defined on a common sample space Ω such that $Var(X)$ and $Var(Y)$ are finite
- ▶ The correlation between X and Y is defined to be

$$\rho_{X,Y} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

Properties of Covariance and Correlation

1. $\text{Cov}(X, c) = 0$ for any X and any constant c
2. $\text{Cov}(X, X) = \text{Var}(X)$ for any X

3.

$$\text{Cov} \left(\sum_{i=1}^n a_i X_i, \sum_{j=1}^m b_j Y_j \right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \text{Cov}(X_i, Y_j)$$

In particular,

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y) \text{ and}$$
$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

Properties of Covariance and Correlation

1. For any two independent random variables X and Y ,
 $\text{Cov}(X, Y) = \rho_{X,Y} = 0$
2. $\rho_{a+bX, c+dY} = \text{sgn}(bd)\rho_{X,Y}$
3. Whenever $\rho_{X,Y}$ is defined, $-1 \leq \rho_{X,Y} \leq 1$
4. $\rho_{X,Y} = 1$ if and only if for some a and $b > 0$
 $P(Y = a + bX) = 1$
5. $\rho_{X,Y} = -1$ if and only if for some a and $b < 0$
 $P(Y = a + bX) = 1$

Some proofs

- ▶ E.g. $\text{Cov}(X, c) = E(cX) - E(c)E(X) = c(EX) - c(EX) = 0$
- ▶ $\text{Cov}(X, X) = E(X^2) - [E(X)]^2 = \text{Var}(X)$
- ▶ E.g. the property (1) follows since $E(XY) = E(X)E(Y)$ if X and Y are independent

Example I: correlation between min and max in dice rolls

- ▶ A fair die is rolled twice; X is the min and Y is the max of two rolls. What is $\rho_{X,Y}$?
- ▶ From the joint distribution of X and Y obtained earlier,

$$E(XY) = 1/36 + 2/18 + 4/36 + \dots = 49/4$$

- ▶ From the marginal pmfs of X and Y , $E(X) = 161/36$, $E(Y) = 91/36$, $Var(X) = Var(Y) = 2555/1296$
- ▶ Thus, $\rho_{X,Y} = \frac{E(XY) - E(X)E(Y)}{\sqrt{Var(X)Var(Y)}} = 0.48$
- ▶ One can show mathematically that this correlation is positive for any number of rolls of a die
- ▶ However, it will converge to zero as the number of rolls tends to infinity

Example II: Correlation does not mean Independence

- ▶ If $\text{Cov}(X, Y) = 0$ X and Y are not necessarily independent
- ▶ Take X such that $P(X = \pm 1) = p$, $P(X = 0) = 1 - 2p$ for some $0 < p < \frac{1}{2}$ and define $Y = X^2$
- ▶ Note that $E(XY) = E(X^3) = 0$; also, since $E(X) = 0$ we have $E(X)E(Y) = 0$
- ▶ Therefore, $\text{Cov}(X, Y) = 0$; however, X and Y are not independent
- ▶ Indeed, note that $P(Y = 0|X = 0) = 1$ but $P(Y = 0) = 1 - 2p \neq 0$
- ▶ More generally, if X has a distribution symmetric around zero and has three finite moments, then X and X^2 always have a zero correlation while not being independent

Definition

- ▶ Let $X_1, X_2, \dots, X_n$ be the discrete random variables defined on a common sample space Ω .
- ▶ Each X_i takes values in some countable set $\aleph_i$
- ▶ The joint pmf of $(X_1, X_2, \dots, X_n)$ is defined as

$$p(x_1, x_2, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$$

whenever $x_i \in \aleph_i$ and 0 otherwise

Requirements of joint pmf

1.

$$p(x_1, \dots, x_n) \geq 0$$

for any $x_1, \dots, x_n \in R$

2.

$$\sum_{x_1 \in \mathcal{X}_1, \dots, x_n \in \mathcal{X}_n} p(x_1, \dots, x_n) = 1$$

Multinomial distribution

- ▶ Let n balls be distributed to k cells independently
- ▶ Each ball has the probability p_i of being dropped into the i th cell
- ▶ We consider X_i , $i = 1, \dots, k$ as the number of balls that gets dropped into the i th cell
- ▶ Their joint pmf with parameters $n, p_1, \dots, p_k$ is

$$P(X_1 = x_1, \dots, X_k = x_k) = \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}$$

where $x_i \geq 0$, $\sum_{i=1}^k x_i = n$, $p_i \geq 0$ and $\sum_{i=1}^k p_i = 1$

- ▶ We write $(X_1, \dots, X_k) \sim \text{Mult}(n, p_1, \dots, p_k)$ - a random vector with a multinomial distribution

Example I

1. Suppose a fair die is rolled 30 times
2. We have $(X_1, \dots, X_6) \sim \text{Mult}(30, p_1, \dots, p_6)$ where each $p_i = \frac{1}{6}$
3. The probability

$$\begin{aligned} P(X_1 = 5, \dots, X_6 = 5) &= \frac{30!}{(5!)^6} \left(\frac{1}{6}\right)^5 \cdots \left(\frac{1}{6}\right)^5 \\ &= 0.0004 \end{aligned}$$

Example I

1. Note that each of the thirty rolls is either six or not
2. The probability of getting 6 on each roll is $\frac{1}{6}$ and the rolls are independent
3. Since $X_6 \sim \text{Bin}(30, 1/6)$,

$$\begin{aligned} P(X_6 \geq 5) &= 1 - P(X_6 \leq 4) = 1 - \sum_{x=0}^4 \binom{30}{x} \left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{30-x} \\ &= .5757 \end{aligned}$$

Basic properties of the multinomial distribution

► Let $(X_1, \dots, X_k) \sim \text{Mult}(n, p_1, \dots, p_k)$. Then

1. $E(X_i) = np_i$; $\text{Var}(X_i) = np_i(1 - p_i)$
2. For any i , $X_i \sim \text{Bin}(n, p_i)$
3. $\text{Cov}(X_i, X_j) = -np_i p_j$ for any $i \neq j$
4. $\rho_{X_i, X_j} = -\sqrt{\frac{p_i p_j}{(1-p_i)(1-p_j)}}$ for any $i \neq j$
5. For any m such that $1 \leq m \leq k$ the conditional distribution

$$(X_1, \dots, X_m) | (X_{m+1} + X_{m+2} + \dots + X_k) = s \sim \text{Mult}(n-s, \theta_1, \dots, \theta_m)$$

$$\text{where } \theta_i = \frac{p_i}{p_1 + \dots + p_m}$$

Basic properties of the multinomial distribution

- ▶ (1) Denote W_{ir} as the indicator of the event that r th ball lands in the i th cell
- ▶ For given i , variables W_{ir} are independent; thus,
$$X_i = \sum_{r=1}^n W_{ir}$$
- ▶ Therefore, $E(X_i) = E(\sum_{r=1}^n W_{ir}) = np_i$ and
$$\text{Var}(X_i) = \text{Var}(\sum_{r=1}^n W_{ir}) = np_i(1 - p_i)$$
- ▶ (2) follows from the definition of a multinomial experiment